anova class contraceptive

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Family health worker receives complaints of weight gain from clients on contraceptives. He has also noticed & is aware that weight gain varies from individual to individual. He wants to find out if it is due to genetic variation or contraceptives. He compares 4 different contraceptive

```
library(readxl)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
anova_contra <- read_excel(here::here("data", "raw", "one_Way_ANOVA classwork_2022.xlsx"))
glimpse(anova_contra)
```

```
## Rows: 24
## Columns: 3
## $ SubjectID     <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
## $ wtgain        <dbl> 12.2, 9.5, 11.6, 13.0, 10.1, 9.6, 4.9, 10.6, 7.0, 8.3, 5~
## $ contraceptive <dbl> 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4,~
```

convert contraceptive to factor

```
anova_contra$contraceptive <- as.factor(anova_contra$contraceptive)
class(anova_contra$contraceptive)
```

```
## [1] "factor"
```

```
levels(anova_contra$contraceptive)
```

```
## [1] "1" "2" "3" "4"
```

```
View(anova_contra)
```

```
#grandmean
mean(anova_contra$wtgain)
```

```
## [1] 8.5
```

```
#find mean in each categogory using dyplyr

anova_contra %>%
  group_by(contraceptive) %>%
  summarise(mean(wtgain))
```

```
## # A tibble: 4 x 2
##   contraceptive `mean(wtgain)`
##   <fct>                  <dbl>
## 1 1                         11
## 2 2                          8
## 3 3                          9
## 4 4                          6
```

```
#please note; the summarise function is different from summary as it only produces the specific calcula
```